THE NINE-COLOR CASE OF AN ERDŐS–GYÁRFÁS BALANCED-COLORING PROBLEM

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Abstract. Erdős and Gyárfás asked whether every r-coloring of $E(K_{r^2+1})$ contains $r+1$ vertices on which some color is absent. We prove the assertion for $r = 9$: every nine-coloring of $E(K_{82})$ has ten vertices whose induced edges omit a color. The proof combines a colored extremal recursion with two terminal results. The first proves $P_3(26) \geq 121$ by exact rational certificates, structural reductions, and solver-free finite-state searches. The second proves that the corresponding family on 27 vertices is empty; a human reduction leaves 50 finite instances, each refuted by a replayed LRAT certificate. A short full-color argument then proves $P_4(37) \geq 192$, which makes every cell of the outer packing recursion strict. The source release contains independent catalog reconstruction, semantic verifiers, corruption tests, manifests, and replay instructions. This fixed case does not settle the problem for arbitrary r.

Verification status. The solver-free proof, full implication chain, independent 101,880-state CaDiCaL reconstruction, and clean public-package replay pass locally. External mathematical review remains pending.

1. Introduction

Erdős and Gyárfás posed the following question [4]. If the edges of K_{r^2+1} are colored with r colors, must some set of $r+1$ vertices omit a color on its induced edges? The assertion is immediate for some small parameters, but no general proof is known. It is listed as Erdős Problem 617.

We establish the nine-color case.

Theorem 1.1. *Every map*

$$\chi : E(K_{82}) \longrightarrow [9]$$

has a set $S \subseteq V(K_{82})$ of order ten such that

$$\chi(E(K_{82}[S])) \neq [9].$$

The argument is by contradiction. For each color i, let G_i contain the edges of color i. If every ten-set sees all colors, then

$$\alpha(G_i) \leq 9 \quad \text{and} \quad e(G_i[S]) \leq 37 \quad (|S| = 10).$$

More importantly, the eight graphs of the other colors partition $E(\overline{G_i})$. This simultaneous decomposition gives induced-density bounds unavailable in a one-color relaxation.

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The proof has three parts.

(i) A colored recursive lower bound turns terminal information into five forced copies of K_9 in a least-color nonneighborhood.
(ii) Two computer-assisted terminal theorems give
$$P_3(26) \geq 121, \qquad \mathcal{P}_3(27) = \varnothing.$$
(iii) A human full-color lemma combines these statements to prove $P_4(37) \geq 192$. The resulting outer margins are $5, 4, 3, 2, 3$ in the only formerly deficient row.

The finite work is not treated as an opaque solver call. The order-26 proof has exact catalog semantics, rational dual checks, and deterministic solver-free searches. The order-27 proof uses LRAT refutations only after a human lemma removes 282 of 332 core types. An independent verifier rebuilds every remaining formula and replays every proof.

2. Colored induced graphs

Assume throughout Sections 2–7 that a counterexample to Theorem 1.1 exists. Fix a target color i. For $n = 9q + b$, $0 \leq b < 9$, put
$$p_9(n) = (9-b)\binom{q}{2} + b\binom{q+1}{2} = 9\binom{q}{2} + qb. \tag{1}$$
This is the minimum edge count in an n-vertex graph with independence number at most nine.

Lemma 2.1 (Full-color induced density)**.** *For every $W \subseteq V(K_{82})$,*
$$e(G_i[W]) \leq D_9(|W|), \qquad D_9(n) := \binom{n}{2} - 8p_9(n).$$

Proof. Each nontarget graph $G_j[W]$ has independence number at most nine, so Turán's theorem gives $e(G_j[W]) \geq p_9(|W|)$. The eight nontarget graphs partition the complement of $G_i[W]$. Summing their lower bounds proves the claim. □

The values used later are
$$D_9(10) = 37, \qquad D_9(11) = 39, \qquad D_9(26) = 125, \qquad D_9(27) = 135. \tag{2}$$

Definition 2.2. Let $\mathcal{F}(a, n)$ be the family of graphs $G_i[W]$ inherited from the same hypothetical coloring, with
$$|W| = n, \qquad \alpha(G_i[W]) \leq a, \qquad \omega(G_i[W]) \leq 8.$$
If the family is nonempty, let $P_a(n)$ be its minimum edge count. We also write $\mathcal{P}_a(n)$ for the family itself and $\mathcal{P}_a(n) = \varnothing$ when it is empty.

The inherited provenance is part of the definition. In particular, every member and every induced subgraph satisfies Lemma 2.1.

2.1. **The colored recursive bound.** We first justify the terminal thresholds used by the recursion.

Lemma 2.3 (Colored layer thresholds)**.** *There is no member of $\mathcal{F}(s,n)$ when*

$$n \geq 9s + T_s, \qquad (T_2, T_3, T_4, T_5, T_6) = (0, 1, 2, 4, 8).$$

Proof. For $s = 2$, let $H \in \mathcal{F}(2,n)$ and put $L = \overline{H}$. The graph L is triangle-free and $\alpha(L) \leq 8$, so $\Delta(L) \leq 8$ and $e(L) \leq 4n$. Lemma 2.1 gives $e(L) \geq 8p_9(n)$. At $n = 18$, equality is forced and L is 8-regular. Since $8 > 2n/5$, the Andrásfai–Erdős–Sós theorem makes L bipartite [1], contradicting $\alpha(L) \leq 8$. For $n > 18$, one has $p_9(n) > n/2$, giving an immediate edge-count contradiction.

Proceed by induction on s. If $H \in \mathcal{F}(s,n)$ and $L = \overline{H}$, then every neighborhood in L induces the complement of a member of the preceding layer. Hence

$$\Delta(L) \leq 9(s-1) + T_{s-1} - 1. \tag{3}$$

Write $n = 9s + t$, $1 \leq t < 9$. For every nontarget color j, the graph $\overline{G_j[W]}$ contains H and is not 9-partite: nine independent sets of H have total order at most $9s < n$. The Kang–Pikhurko theorem therefore gives

$$e(G_j[W]) \geq p_9(n) + s - 1.$$

Summing over the eight nontarget colors and comparing with (3) gives a contradiction exactly when

$$A_s t - \widehat{B}_s > 0,$$

where

$$A_s = 9(s+1) - 2s - T_{s-1} + 1, \qquad \widehat{B}_s = 9s(s + T_{s-1} - 2) - 16(s-1).$$

Thus the first eligible offset is

$$T_s = \max\left\{1, \left\lfloor \frac{\widehat{B}_s}{A_s} \right\rfloor + 1\right\},$$

provided it is less than nine. Starting from $T_2 = 0$ gives $T_3 = 1$, $T_4 = 2$, $T_5 = 4$, and $T_6 = 8$. The edge-count margin increases at every later order: its doubled one-vertex increment is at least

$$16\left\lfloor \frac{n}{9} \right\rfloor - (9(s-1) + T_{s-1} - 1) \geq A_s > 0.$$

Thus each contradiction persists beyond its first offset. $\square$

We record the exact recursion used in the final calculation. For an independence cap a, define the Turán floor

$$p_a(n) = (a-b)\binom{q}{2} + b\binom{q+1}{2}, \qquad n = aq + b, \quad 0 \leq b < a.$$

Set

$$B(0,0) = 0, \qquad B(0,n) = +\infty \ (n > 0),$$

and

$$B(1,n) = \begin{cases} \binom{n}{2}, & n \leq 8, \\ +\infty, & n \geq 9. \end{cases}$$

For $a \geq 2$, put

$$Q(a,n) = p_a(n) + \begin{cases} 0, & n \leq 8a, \\ \lfloor n/a \rfloor - 1, & 8a < n \leq 9a, \\ \lfloor n/a \rfloor, & n \geq 9a+1, \end{cases} \tag{4}$$

and

$$C(d) = \begin{cases} \binom{d+1}{2}, & d < 8, \\ 37, & d = 8, \\ d + \left\lceil \frac{11}{9}\binom{d}{2} \right\rceil, & d \geq 9. \end{cases} \tag{5}$$

By Lemma 2.3, the exact colored-layer thresholds used here are

$$T_2 = 0, \quad T_3 = 1, \quad T_4 = 2, \quad T_5 = 4, \quad T_6 = 8. \tag{5a}$$

Whenever T_a is defined and $n \geq 9a + T_a$, set $B(a,n) = +\infty$. Otherwise set

$$R(a,n) = \min_{\substack{0 \leq d < n \\ B(a-1,n-1-d) < +\infty}} \max\left\{ B(a-1, n-1-d) + C(d), \left\lceil \frac{nd}{2} \right\rceil \right\}, \tag{6}$$

then put $b(a,n) = \max\{Q(a,n), R(a,n)\}$. If $b(a,n) > D_9(n)$, set $B(a,n) = +\infty$; otherwise set $B(a,n) = b(a,n)$.

Proposition 2.4 (Recursive lower bound)**.** *Every $F \in \mathcal{F}(a,n)$ satisfies $e(F) \geq B(a,n)$. A value $B(a,n) = +\infty$ certifies that $\mathcal{F}(a,n)$ is empty.*

Proof. Induct on a. The cases $a = 0, 1$ follow from the definitions. The scalar bound (4) is Turán's bound plus the Kang–Pikhurko nonpartite increment [5]. The complement of F is not a-partite when $n > 8a$, since one part would be a target clique of order at least nine. To justify the last extra unit, consider equality in their construction. If $n \geq 9a + 2$, its a old parts have total order $n - 1 > 9a$, so one gives a target clique of order at least ten. If $n = 9a + 1$, every old part has order nine. For the special old part S, exceptional vertices x, y, and nonempty proper set $A \subset S$, the two ten-sets $S \cup \{x\}$ and $S \cup \{y\}$ have respectively

$$\binom{9}{2} + 9 - |A| \quad \text{and} \quad \binom{9}{2} + |A|$$

target edges. The cap 37 would require both $|A| \geq 8$ and $|A| \leq 1$, a contradiction. This proves (4).

Choose a minimum-degree vertex x of F, of degree d, and let U be its nonneighbor set. Then $F[U] \in \mathcal{F}(a-1, n-1-d)$. If $X = e_F(N(x), U)$ and $Y = e_F(N(x))$, minimum degree gives

$$X + 2Y \geq d(d-1).$$

For $d < 8$ this yields $d + X + Y \geq \binom{d+1}{2}$. At $d = 8$, equality would produce a target K_9, giving one additional edge. For $d \geq 9$, averaging the ten-set cap over the nine-subsets of $N(x)$ gives the last line of (5). The degree-sum bound also gives $e(F) \geq \lceil nd/2 \rceil$. These two estimates yield (6). Lemma 2.1 justifies the final emptiness test, and Lemma 2.3 justifies the declared terminal values. ∎

The terminal results proved below strengthen this recursion by setting

$$B(3,26) \geq 121, \qquad B(3,27) = +\infty. \tag{7}$$

A third input, proved in Section 6, is

$$B(4,37) \geq 192. \tag{8}$$

3. The packing contradiction

Lemma 3.1 (Representative selection)**.** *Let Q be a target copy of K_9 and let $x \notin Q$. At most one edge from x to Q has the target color. Consequently, from k disjoint target copies of K_9 one can choose representatives target-anticomplete to any fixed target-independent set I, provided $|I| + k \leq 9$.*

Proof. The ten-set $Q \cup \{x\}$ must see all eight nontarget colors. Those colors can occur only on the nine edges from x to Q, so at most one of those edges has the target color. Choose representatives successively. Before choosing from the next block, at most eight earlier vertices are present; each excludes at most one of its nine vertices. At least one choice remains. □

Let G be a color graph with the fewest edges. Then

$$e(G) \leq \frac{1}{9}\binom{82}{2} = 369. \tag{9}$$

Its minimum degree is at most eight. Indeed, minimum degree at least nine would force G to be nine-regular. Since $\alpha(G) \leq 9$, some component has chromatic number at least ten, contradicting Brooks's theorem unless it is K_{10} [2]. A target K_{10} contradicts the ten-set cap.

Fix a minimum-degree vertex v of degree $d \leq 8$, and let U be its target nonneighbor set. Suppose j disjoint target copies of K_9 have been selected in U, and let R_j be the residual. If R_j has no further K_9, representative selection gives

$$R_j \in \mathcal{F}(8-j, 81-d-9j). \tag{10}$$

To see the independence bound, an independent $(9-j)$-set in R_j could be extended by one representative from each selected block using Lemma 3.1. The resulting independent nine-set in U, together with v, would contradict $\alpha(G) \leq 9$.

Put $N = N_G(v)$, $X = e_G(N,U)$, and $Y = e(G[N])$. Summing the degree lower bound over N gives

$$X + 2Y \geq d(d-1).$$

Since $Y \leq \binom{d}{2}$, it follows that $X + Y \geq \binom{d}{2}$. The least-color budget therefore gives

$$e(R_j) \leq E_{d,j} := 369 - d - \binom{d}{2} - 36j. \tag{11}$$

Therefore

$$B(8-j, 81-d-9j) > E_{d,j} \tag{12}$$

forces the packing to extend by one block.

Lemma 3.2 (Five-block exclusion)**.** *The graph $G[U]$ cannot contain five disjoint copies of K_9.*

Proof. Extend five blocks to a maximal packing of k blocks and put $s = 8-k$. The same representative argument gives independence number at most s in the residual, whose order is

$$9s + (9 - d).$$

For $k = 5$, this contradicts the colored threshold $T_3 = 1$. For $k = 6$, the threshold $T_2 = 0$ applies. If $k = 7$, the residual has independence number at most one and order at least ten, so it is a target clique, contrary to maximality. If $k \geq 8$, eight blocks leave at least one vertex of U outside them. Lemma 3.1 selects one representative from each block target-anticomplete to that vertex. These nine vertices form an independent set in $G[U]$, and adding v gives an independent ten-set in G. Thus no maximal extension of five blocks exists. □

4. The 26-vertex terminal

Theorem 4.1. *Every $H \in \mathcal{F}(3, 26)$ has at least 121 edges.*

Proof. Suppose $e(H) \leq 120$. A vertex of degree at most seven has at least 18 target nonneighbors. Their induced graph belongs to $\mathcal{F}(2, n)$ for some $n \geq 18$, contrary to Lemma 2.3. Hence $\delta(H) \geq 8$, while averaging gives $\delta(H) \leq 9$.

The degree-eight branch is excluded by Lemma 4.2, proved below. In the degree-nine branch choose a degree-nine vertex v and put

$$A = N_H(v), \quad B = V(H) \setminus (A \cup \{v\}), \quad F = \overline{H[A]}, \quad L = \overline{H[B]}.$$

Thus $|A| = 9$ and $|B| = 16$. The graph L is triangle-free, so $e(L) \leq 64$. Lemma 2.1 on B gives $e(L) \geq 8p_9(16) = 56$. Hence

$$56 \leq e(L) \leq 64.$$

Proposition 4.3 excludes every one of these nine levels. □

4.1. **A common human exclusion.**

Lemma 4.2 (Seventeen-vertex two-row exclusion)**.** *Neither of the following configurations exists:*

(i) $|V(H)| = 26$, $\delta(H) = 8$, *and* $e(H) \leq 120$*;*
(ii) $|V(H)| = 27$, $\delta(H) = 9$, *and* $e(H) \leq 135$*,*

where $H \in \mathcal{F}(3, |V(H)|)$.

Proof. Choose a minimum-degree vertex v, put $A = N_H(v)$ and $B = V(H) \setminus (A \cup \{v\})$, and set $F = \overline{H[A]}$, $L = \overline{H[B]}$. In both cases $|B| = 17$. The checked core catalog has fourteen types. Every type has a vertex z such that $L - z$ is bipartite with sides X, Y of order eight and $d_L(z) = p + q \leq 5$, where p and q count its neighbors in X and Y. Both are positive. If h is the number of missing X–Y edges, then

$$h \leq p + q. \tag{13}$$

The graph F has an edge cu. Put $S = (N_H(c) \cap B) \cup (N_H(u) \cap B)$. This is a vertex cover of L; otherwise c, u and an uncovered edge of L form an independent four-set in H. Applying $D_9(11) = 39$ to $\{c, u\} \cup X \cup \{z\}$ and to the analogous set using Y gives

$$|S \cap X| + \mathbf{1}_{z\in S} \le p + 3, \qquad |S \cap Y| + \mathbf{1}_{z\in S} \le q + 3.$$

If $z \in S$, the independent set $B \setminus S$ forces

$$h \ge (6 - p)(6 - q) > p + q.$$

If $z \notin S$, the missing neighborhood rectangle and the remaining independent cross-pairs are disjoint, giving

$$h \ge pq + (5 - p)(5 - q) > p + q.$$

Both inequalities contradict (13). The finite dependency is only the fourteen-core catalog and its displayed witness. □

4.2. **The nine core levels.** For $a \in A$, write

$$D_a = N_H(a) \cap B, \qquad q_a = |D_a|.$$

The finite reduction uses the following necessary conditions:

(C1) L is triangle-free, $\alpha(L) \le 8$, and $\Delta(L) \le 8$;
(C2) F is K_4-free and $\alpha(F) \le 7$;
(C3) minimum degree gives $q_a \ge d_F(a)$ and column demands

$$\sum_{a\in A} \mathbf{1}_{b\in D_a} \ge d_L(b) - 6;$$

(C4) if $aa' \in E(F)$, then $D_a \cup D_{a'}$ covers L;
(C5) if a, a', a'' form a triangle of F, then $D_a \cup D_{a'} \cup D_{a''} = B$;
(C6) no row D_a contains an independent eight-set of L;
(C7) the target-edge ledger bounds $\sum_a q_a$ at the relevant level.

Each condition follows from minimum degree, the exclusion of independent four-sets, or the exclusion of target K_9.

Proposition 4.3 (Finite order-26 exclusion)**.** *No incidence system satisfying* (C1)–(C7) *exists at any core level* $e(L) = 56, \ldots, 64$.

Proof. Canonical generation lists every eligible core and shell up to isomorphism. Exact rational duals remove most core–shell pairs. For a retained dual, every incidence variable has total coefficient at most one, while the weighted right side exceeds the global incidence budget. All coefficients are checked as rational numbers.

The remaining pairs are reduced to integral row-size and side-count states. The production searches are deterministic and solver-free. For $m = 58, \ldots, 64$, the core is $K_{8,8}$ with $64 - m$ cross-edges deleted. Its two sides control the row domains and core-cover conditions. Table 1 gives the proof-facing accounting.

At $m = 63$, exact duals leave 334 pairs, scalar conditions leave thirteen, and the orbit recurrence rejects all 10,715 surviving row states. At $m = 64$, exact duals leave 562 pairs from 7,454. The scalar replay checks 101,880 states and leaves 10,639. A human three-hub argument excludes four states. In each, the three hubs c, u, w form a path with center c. Writing the sides of the core $K_{8,8}$ as X, Y, the scalar assignments give, up to exchanging the sides,

$$|D_c \cap X| = 1, \quad |D_c \cap Y| = 6, \quad D_u, D_w \subset X, \quad |D_u| = |D_w| = 7.$$

TABLE 1. Order-26, degree-nine proof ledger.

m	cores	shells	primary final exclusion	independent audit
56,57	431	18	minimum-cover and support classification	6,804 support pairs reconstructed
58	50	55	row states and final shell-triangle lemma	5,056 Boolean states
59	20	85	two-hub finite recurrence	11,943 UNSAT
60	10	152	two-hub recurrence and $K_{5,3} + K_1$ lemma	23,157 UNSAT
61	4	393	side counts and set-system recurrence	29,155 UNSAT
62	2	1,147	side counts and orbit recurrence	38,342 UNSAT
63	1	2,245	cover-13 lemma and orbit recurrence	44,504 UNSAT
64	1	7,454	three-hub lemma and orbit recurrence	101,880 CaDiCaL UNSAT

If $D_c \cap X = \{x_0\}$ and $S = D_c \cap Y$, the two hub-edge cover conditions force

$$D_u = D_w = X \setminus \{x_0\}.$$

Every row adjacent to all three hubs then equals $(Y \setminus S) \cup \{x_0\}$. The six vertices of S still need a second incidence, but the four states have only $4, 0, 1, 2$ remaining row positions. This is impossible.

A solver-free $S_8 \times S_8$ orbit recurrence excludes the other 10,635 states after 3,883,026 orbit representatives and 299,684,034 demand states. Thus every generated pair is rejected. Independent Boolean reconstruction is an audit of this last conclusion, not its proof premise. □

5. The 27-vertex terminal

Theorem 5.1. *The family $\mathcal{P}_3(27)$ is empty.*

Proof. Let $H \in \mathcal{F}(3, 27)$. Lemma 2.1 gives $e(H) \leq 135$. A vertex of degree at most eight has at least 18 target nonneighbors, which induce a forbidden member of the two-layer family. Hence $\delta(H) \geq 9$, while averaging gives $\delta(H) \leq 10$.

Lemma 4.2 excludes $\delta(H) = 9$. If $\delta(H) = 10$, then H is 10-regular and has 135 edges. Proposition 5.2 excludes this branch. □

Proposition 5.2 (Regular endpoint)**.** *There is no 10-regular graph $H \in \mathcal{F}(3, 27)$.*

Proof. Choose $v \in V(H)$ and put

$$A = N_H(v), \quad B = V(H) \setminus (A \cup \{v\}), \quad F = \overline{H[A]}, \quad L = \overline{H[B]}.$$

Thus $|A| = 10$, $|B| = 16$. For $b \in B$, let $C_b = N_H(b) \cap A$. Regularity gives

$$|C_b| = d_L(b) - 5. \tag{14}$$

The graph F has independence number at most seven, so $\tau(F) \geq 3$. For every edge $bb' \in E(L)$, the union $C_b \cup C_{b'}$ covers F; otherwise an uncovered edge of F and bb' form an independent four-set in H. Consequently

$$d_L(b) + d_L(b') \geq 13 \qquad (bb' \in E(L)). \tag{15}$$

Independent canonical generation produces 332 eligible core types, with edge-level counts

$$179, 80, 39, 17, 9, 4, 2, 1, 1$$

for $e(L) = 56, \dots, 64$. The minimum endpoint-degree-sum profile is

minimum sum	10	11	12	13	14	15	16
cores	19	102	161	38	10	1	1.

Thus (15) rejects 282 cores and leaves 50.

For each survivor, a deterministic CNF encodes a relaxation of the configuration. It contains exact degrees and edge counts, every split of an independent four-set, every mixed target K_9, the full-color 11-set cap, and the local deletion inequalities. Constraints omitted from the formula only make it weaker, so an actual configuration would satisfy the CNF.

The 50 formulas are UNSAT. Their LRAT refutations are independently replayed. The semantic verifier reconstructs all 332 cores, recomputes the 50 indices, rebuilds each formula without importing the generator, compares clause multisets, checks every unary totalizer shape, and replays every LRAT proof. Its aggregate receipt is

```
cases_reconstructed=332
cnfs_verified=50
clauses_verified=9100515
totalizer_merges_verified=79574
totalizer_shapes_verified=83
totalizer_states_verified=108839
lrat_proofs_replayed=50
status=PASS
```

This excludes all 50 survivors and proves the proposition. □

6. The full-color bridge

The terminal statements interact more strongly than scalar averaging suggests.

Lemma 6.1. *Every $H \in \mathcal{F}(4, 37)$ has at least 192 edges.*

Proof. Suppose $e(H) \le 191$. If a vertex had degree at most nine, at least 27 of its nonneighbors would induce a member of $\mathcal{F}(3, 27)$, contrary to Theorem 5.1. Hence $\delta(H) \ge 10$. Since $2e(H)/37 < 11$, there is a degree-ten vertex v.

Put

$$A = N_H(v), \qquad B = V(H) \setminus (A \cup \{v\}), \qquad M = 45 - e(H[A]).$$

Thus $|A| = 10$ and $|B| = 26$. The 11-set $\{v\} \cup A$ has $10 + 45 - M$ target edges. By $D_9(11) = 39$,

$$M \ge 16. \tag{16}$$

For $a \in A$, put $D_a = N_H(a) \cap B$. If $F = \overline{H[A]}$, minimum degree gives

$$10 \le d_H(a) = 1 + 9 - d_F(a) + |D_a|,$$

so $|D_a| \geq d_F(a)$. Therefore

$$e_H(A,B) = \sum_{a\in A} |D_a| \geq 2M. \tag{17}$$

The graph $H[B]$ belongs to $\mathcal{F}(3,26)$ because v is target-anticomplete to B. Theorem 4.1 gives $e(H[B]) \geq 121$. Keeping the missing-edge variable throughout,

$$\begin{aligned} e(H) &= 10 + (45 - M) + e_H(A,B) + e(H[B]) \\ &\geq 10 + (45 - M) + 2M + 121 \\ &= 176 + M \\ &\geq 192, \end{aligned}$$

contrary to the assumption. This proves the lemma. □

Remark 6.2. It is invalid to replace the last display by $10 + 29 + 32 + 121$: the inequality $e(H[A]) \leq 29$ points in the wrong direction for a lower bound. The variable M is needed because the cross-edge gain $2M$ compensates for the shell term $45 - M$.

7. Completion of the proof

Insert Theorems 4.1 and 5.1 and Lemma 6.1 into Proposition 2.4. Along the degree-eight diagonal one obtains

$$\begin{array}{c|ccccc} (a,n) & (4,37) & (5,46) & (6,55) & (7,64) & (8,73) \\ \hline B(a,n) & 192 & 227 & 264 & 301 & 338. \end{array} \tag{18}$$

Table 2 gives all outer packing margins. The symbol $+\infty$ denotes a certified empty residual family.

Table 2. Margins $B(8-j, 81-d-9j) - E_{d,j}$.

d	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$
0	79	82	$+\infty$	$+\infty$	$+\infty$
1	66	69	$+\infty$	$+\infty$	$+\infty$
2	56	55	72	$+\infty$	$+\infty$
3	43	42	45	$+\infty$	$+\infty$
4	35	34	35	$+\infty$	$+\infty$
5	24	23	22	$+\infty$	$+\infty$
6	18	17	16	17	$+\infty$
7	9	8	7	6	$+\infty$
8	5	4	3	2	3

Proof of Theorem 1.1. Choose a least color graph G and a minimum-degree vertex v. By (9), its degree d lies in $\{0,\ldots,8\}$. Every entry of the corresponding row in Table 2 is positive or certifies an empty no-next-block family. Applying (12) successively forces five disjoint target copies of K_9 in the nonneighbor set of v. This contradicts Lemma 3.2. Therefore the assumed coloring does not exist. □

8. Certificate semantics and trust boundary

The proof has four distinct verification layers.

V1. **Human reduction.** Sections 2–7 connect an arbitrary coloring to finite graph and incidence families.

V2. **Catalog completeness.** Canonical graph streams are generated with nauty's `geng` [6]. Independent programs check every graph property, count, and graph6 digest used in the paper.

V3. **Finite exclusion.** Rational duals are checked over exact fractions. Solver-free searches enumerate declared states and use deterministic recurrences. The order-27 branch uses checked LRAT refutations [3].

V4. **Independent reconstruction.** Separate source rebuilds the Boolean semantics, compares generated clauses as multisets, and checks the catalog classifications. These programs do not treat solver logs as proofs.

The order-27 certificate is a relaxation: it omits some consequences of the full coloring. This direction is sound because every genuine configuration satisfies the retained clauses. UNSAT excludes a genuine configuration; SAT would not by itself construct a coloring.

The order-26, level-64 theorem is solver-free. Its independent CaDiCaL run checks the same 101,880 raw row-size states using a separately written Boolean encoding and incremental totalizers. It returns 101,880 UNSAT, zero SAT, and zero UNKNOWN, with classification SHA-256 `e68dfcaf336ff575e0091a757e60874b81cd79966b418df2e6487eb698ade331`. This audit is deliberately not a premise of Proposition 4.3.

9. Reproducibility and data availability

The release repository contains:

- all theorem notes and verifier source;
- graph and shell generators with pinned catalog hashes;
- exact rational-dual data and semantic checkers;
- deterministic solver-free replay programs;
- four order-27 LRAT shards with manifests and checksums;
- corruption tests and a single dependency-ordered replay entry point;
- the source and PDF of this paper.

The public repository location is

`https://github.com/Robby955/erdos-617-fixed-cases`.

The fixed-$r = 9$ files are under `r9/`. The four large certificate archives are attached to release tag `fixed-r9-2026-07-21`. Their names, sizes, and SHA-256 values are recorded in `r9/ASSET_INDEX.json`. The dependency-ordered release replay passed from the exact public directory layout.

The order-27 package currently has four manifest SHA-256 values:

```
f332adb9327c02b2b9964a1d781baae75d27371a83995dc3c1f7393b35c09cba
0df642bb21e52310ec19f479d5ec62fe179a0b075738643bb165baeef7dcda40
7427d8c307411374f332056bdd3638ff26c9d70c6c1283458d87f6313fd1bc2f
e1cc2b4b6f2643d16bc3ba64d50e7e1e9cf73d7043a19f632571bf328c158d1d
```

The release verifier also rejects modified manifests, same-size formula mutations, truncated formulas, false source hashes, and truncated LRAT proofs.

10. AI-use disclosure

OpenAI GPT 5.6 Sol was used substantially for proof exploration, intermediate counterexample searches, adversarial criticism, verification programs, audit planning, research-note organization, and manuscript drafting. This use was not limited to copy editing.

Model agreement is not treated as mathematical review. The retained claims are tied to explicit semantics, exact calculations, deterministic finite searches or replayed proof certificates, independent reconstructions, hashes, and corruption tests. The author selected the proof program, decided which arguments and computations were retained, checked the final implication chain, and assumes responsibility for every claim and error.

11. Scope and nonclaims

Theorem 1.1 is a fixed-parameter result. It does not imply the case $r = 10$ and does not settle Erdős Problem 617 for arbitrary r. No Lean formalization is claimed. At the date on the title page, the preprint has not received external mathematical review.

Appendix A. Dependency map

Table 3. Logical and computational dependencies.

result	human dependency	finite dependency
$P_3(26) \geq 121$	degree split; core–shell incidence lemmas	core levels 56–64, exact duals, solver-free replays
$\mathcal{P}_3(27) = \varnothing$	two-row lemma; regularity; degree-sum 13	332-core catalog and 50 LRAT refutations
$P_4(37) \geq 192$	Theorems 4.1, 5.1; the M-variable argument	integer arithmetic verifier
all 45 outer cells	recursive bound; representative selection	exact integer recurrence replay
Theorem 1.1	least-color reduction; five-block exclusion	the preceding dependency closure

Canonical proof notes and replay commands are listed in `output/ERDOS_617_R9_RELEASE_HANDOFF.md`. That handoff records the exact release commits, artifact locations, tool versions, and nonclaims. Historical checkpoint notes are not authoritative after the release commit.

References

1. Béla Andrásfai, Paul Erdős, and Vera T. Sós, *On the connection between chromatic number, maximal clique and minimal degree of a graph*, Discrete Mathematics **8** (1974), no. 3, 205–218.
2. R. L. Brooks, *On colouring the nodes of a network*, Mathematical Proceedings of the Cambridge Philosophical Society **37** (1941), no. 2, 194–197.
3. Luís Cruz-Filipe, Marijn J. H. Heule, Warren A. Hunt, Jr., Matt Kaufmann, and Peter Schneider-Kamp, *Efficient certified RAT verification*, Automated Deduction – CADE 26, Lecture Notes in Computer Science, vol. 10395, Springer, 2017, pp. 220–236.
4. Paul Erdős and András Gyárfás, *Split and balanced colorings of complete graphs*, Discrete Mathematics **200** (1999), no. 1–3, 79–86.
5. Mihyun Kang and Oleg Pikhurko, *Maximum K_{r+1}-free graphs which are not r-partite*, Matematychni Studii **24** (2005), no. 1, 12–20.
6. Brendan D. McKay and Adolfo Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation **60** (2014), 94–112.